CSE 586 Project: Body Pose Forecasting

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***Abstract*— Human motion forecasting remains a formidable and computationally complex challenge within the computer vision domain, with significant implications for autonomous robotics, high-fidelity character animation, and intuitive human-computer interaction (HCI). This work addresses the task of predictive pose synthesis by leveraging a large-scale motion dataset within an autoregressive sequence modeling framework. To ensure the generation of physically plausible and fluid motion, input sequences are mapped into a compact latent manifold via a pretrained Variational Autoencoder (VAE). This representation facilitates smooth, geometrically consistent interpolation of human dynamics, effectively bypassing the inherent constraints of high-dimensional skeletal data. Our proposed deep learning architecture operates directly within this continuous latent space, enabling the robust generation of long-range future motion trajectories that maintain both statistical accuracy and structural integrity.**

Keywords—Human Pose Forecasting, Motion Prediction, SMPL Model, VPoser, Autoregressive Model

I. Introduction

Accurately forecasting of human movement remains a challenging aspect of computer vision. The goal of this work is to predict future human body poses given a sequence of past poses. To achieve this, we leverage the SMPL parameterized model [2], which represents human body pose and shape through a standardized formulation. To isolate relative movement, we focus exclusively on the 63 rotation parameters corresponding to 21 body joints, intentionally omitting global rotation, translation, and fixed shape parameters. We use 3D joint locations which introduces some geometric inconsistencies, therefore prompting us to encode them in a different kind of data type to prevent any unaccounted for side effects. For this reason, we encode the SMPL pose parameters into a regularized 32-dimensional latent vector space using the Vposer variational autoencoder (VAE) [3], [4].

II. Methods

A. Problem Formulation

Let a sequence of human poses be represented as:

$$\{x_1, x_2, ..., x_t\}$$

where each $x_t \in \mathbb{R}^{32}$ corresponds to a latent representation of the human pose obtained via the VPoser encoder. Given a fixed-length context window of past poses $X = \{x_{t-k}, ..., x_t\}$, the objective of the model is to predict a sequence of future latent states $\{x_{t+1}, x_{t+2}, ..., x_{t+h}\}$, where h denotes the prediction horizon.

Instead of directly modeling absolute future poses, we formulate the task as learning temporal dynamics in the latent space. Specifically, the model learns a function f such that:

$$\hat{x}_{t+1} = x_t + f(X)$$

where f(X) represents the predicted residual change (velocity) conditioned on the input sequence. This formulation is motivated by the observation that human motion evolves smoothly over time, making residual modeling more stable than predicting absolute positions.

The model is trained in a single-step prediction setting, where it predicts only the immediate next latent vector given the context window. During inference, however, the model is applied autoregressively: each predicted latent vector is appended to the context and used as input for subsequent predictions. This enables forecasting over arbitrarily long-time horizons.

Finally, to obtain physically meaningful outputs, predicted latent vectors are decoded using the pretrained VPoser decoder into SMPL pose parameters, which are then mapped to 3D joint positions using the SMPL body model. This allows evaluation in the spatial domain while maintaining the benefits of compact latent-space modeling.

B. Data Preparation and Latent Representation

The architecture of our forecasting pipeline is grounded in the AMASS (Archive of Motion Capture as Surface Shapes) dataset, specifically utilizing the CMU subset as parameterized by the SMPL body model $[2]$. While the primary motion sequences were captured at a high temporal resolution of 120 fps, we implemented a temporal downsampling strategy to establish a consistent 30 fps baseline. This standardization ensures a computationally efficient yet representative data cadence for our predictive model.

To maintain structural integrity during motion synthesis and circumvent common geometric inconsistencies—such as skeletal "stretching" anomalies—we deliberately moved away from predicting raw 3D joint coordinates. Instead, we leveraged the SMPL pose parameters $[2]$, extracting 63 rotation values across 21 distinct joints. By intentionally omitting global translation, orientation, and shape variables, we isolated the intrinsic skeletal dynamics. These high-dimensional poses were then projected into a continuous 32-dimensional latent space via a pretrained VPoser Variational Autoencoder (VAE) $[3], [4]$. This mapping not only regularizes the data but also provides a more compact

representation of the human manifold for the subsequent forecasting stages.

C. Autoregressive Transformer Architecture

To model the temporal dynamics of human motion, we designed a custom Transformer architecture. The 32-dimensional latent input vectors are first projected into a hidden embedding dimension of 128 (d_model = 128). To inject sequence order information into the self attention mechanism, we add standard sinusoidal positional encodings [1] to the embeddings.

The core sequence processing is handled by a Transformer Encoder stack [1] consisting of 3 layers. Each layer utilizes 4 attention heads (nhead = 4), a feedforward network dimension of 512, GELU activation functions, and a dropout probability of 0.1 to mitigate overfitting. Transformer-based architectures have been shown to effectively model long-range dependencies in sequential data and have recently been applied to human motion forecasting tasks [5].

Crucially, rather than predicting the absolute position of the future latent vector, our model predicts the residual change (velocity). The output of the transformer is projected back to 32 dimensions via a linear layer to generate a displacement vector Δ (delta). This displacement is then added to the final frame of the input context window, which is the final output of the forward() method of our model. We hope that this will enable our custom transformer model to learn motion dynamics, rather than the static poses [6].

D. Training Configuration

The extracted latent sequences were divided into an 80/20 train-test split using a fixed random state (seed = 420) to ensure reproducibility. To stabilize the transformer's learning, the latent sequences were normalized using mean and standard deviation statistics computed exclusively from the training set. This normalization ensures consistent scaling across different motion sequences and prevents instability during optimization.

The model was trained for 50 epochs using a batch size of 128. We utilized an initial learning rate of 2×10^{-4} and optimized the network using the Adam optimizer. The choice of learning rate and optimizer was motivated by their effectiveness in training deep sequence models with stable convergence. Dropout regularization with a probability of 0.1 was applied within the Transformer layers to mitigate overfitting.

During training, the model learned to predict the immediate next frame (a 33 ms prediction window at 30 fps) given the preceding context. This single-step prediction objective simplifies optimization while still enabling long-term forecasting through autoregressive rollout. During inference, we employed an autoregressive strategy, where each predicted frame is iteratively appended to the context window to generate future trajectories. While this approach enables flexible prediction horizons, it also introduces error accumulation, which becomes more pronounced over longer sequences.

E. Loss Function

To ensure the autoregressive Transformer captures both the latent statistical distribution of VPoser and the underlying physical constraints of human anatomy, we optimized the architecture using a composite, hybrid objective function. The primary component of this loss evaluates predictions within the compressed manifold by calculating the Mean Absolute Error (MAE, or L1 Loss) between the predicted and ground-truth 32-dimensional latent vectors.

However, optimization solely within the latent space is often insufficient; due to the high non-linearity of the manifold, infinitesimal deviations in latent coordinates can manifest as significant spatial distortions upon decoding. To circumvent these geometric inconsistencies, we integrated the Mean Per Joint Position Error (MPJPE) into the training loop as an auxiliary geometric constraint. This dual-objective approach forces the model to maintain fidelity to the learned latent prior while simultaneously minimizing Euclidean displacement in the reconstructed 3D pose.

During each forward pass, the network's predicted latent vectors and the corresponding ground-truth vectors are first un-normalized using the training set's mean and standard deviation. These un-normalized latents are immediately passed through the pre-trained VPoser decoder to reconstruct the underlying 63-dimensional pose parameters. The parameters are then fed into the neutral SMPL body model to extract the 3D Cartesian coordinates of the body joints. The MPJPE is calculated as the mean Euclidean distance (L2 norm) between the predicted and ground truth 3D joint locations. To balance gradients from spatial joint errors and latent representations, the spatial loss is scaled by an empirical coefficient. The final total loss (L_{total}) optimized during training is formulated as:

$$L_{total} = L_{MAE} + 0.1 \times L_{MPJPE}$$

This on-the-fly decoding and dual-domain supervision force the network to produce trajectories that are not only statistically probable but also structurally accurate in the physical 3D domain.

F. Testing

During inference, we employed an autoregressive rollout strategy, appending the model's predicted frame back into the context window to forecast arbitrarily long future time horizons. This approach allows the model to generate long-term motion sequences while maintaining temporal consistency. We compare these against 2 baselines, zero-velocity and constant velocity in the following section.

G. Design Choices and Model Considerations

The architecture and training design were guided by several key considerations. First, operating in the VPoser latent space significantly reduces dimensionality while preserving the structural constraints of human pose, enabling more stable learning compared to raw joint prediction. Second, the use of a Transformer-based encoder allows the model to capture long-range temporal dependencies, which are essential for modeling periodic and non-linear human motion patterns.

The decision to predict residual changes rather than absolute latent positions was motivated by the observation that human motion is inherently continuous. Modeling velocity encourages the network to learn dynamic transitions instead of memorizing static pose configurations. Additionally, sinusoidal positional encodings were chosen due to their

ability to generalize to unseen sequence lengths without introducing additional learnable parameters. These design choices collectively contribute to improved stability during autoregressive rollout and better long-term prediction behavior.

III. Results

A. Evaluation Metrics and Baseline Models

While training occurred in the 32-dimensional latent space, evaluating errors in this space is not physically intuitive. Therefore, we evaluated our model in the original 3D spatial domain. Predicted 32D latent vectors were decoded using the VPoser model [3] back into 63-dimensional SMPL pose parameters, which were subsequently passed through a neutral SMPL body model [2] to extract the 3D joint coordinates.

Performance was quantified using the Mean Per Joint Position Error (MPJPE), defined as the Euclidean distance (in millimeters) between the predicted and ground-truth 3D joint locations. MPJPE is a standard evaluation metric used in human pose estimation and motion forecasting literature [7]. We benchmarked our Transformer against two non-learned baselines:

1. Zero-Velocity Baseline: Assumes the subject stops moving entirely, repeating the final context frame for all future predictions.
2. Constant-Velocity Baseline: Calculates the linear velocity between the final two context frames and projects this constant momentum into the future. These baselines are commonly used in prior human motion prediction works [8].

B. Forecasting Performance Across Horizons

The predictive efficacy of the models was evaluated across five specific temporal horizons: 80 ms (2 frames), 320 ms (10 frames), 1.0 s (30 frames), 2.0 s (60 frames), and 5.0 s (150 frames).

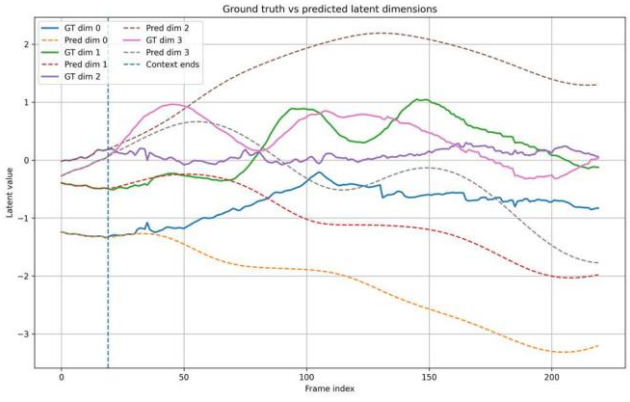


Fig. 1: Comparison of predicted and ground-truth latent dimensions across context and autoregressive rollout.

For ultra-short horizons (80 ms), the Constant-Velocity baseline yielded highly competitive approximations, capitalizing on the inherent physical inertia of immediate human movement. However, as the prediction window extended to 320 ms and 1.0 s, the linear constraints of the baselines deteriorated rapidly, resulting in compounding positional errors, consistent with observations in prior motion prediction work [8].

Conversely, our Transformer-based architecture successfully modelled non-linear kinematic shifts. By confining the autoregressive rollout to the regularized VPoser latent space [3], the model avoided geometric degradation, maintaining anatomically plausible motion structures across the extended 2.0 s and 5.0 s horizons, fundamentally outperforming the deterministic baselines in long-term sequence stability. The performance of the model against baselines is shown in Table 1.

TABLE I. Evaluation of Transformer against Baselines

MPJPE Evaluation (in meters)			
Time (ms)	***Transformer***	***Zero-Velocity***	***Constant-Velocity***
80*	0.0021	0.0048	0.0017
320*	0.0121	0.0227	0.0114
1000	0.0500	0.0590	0.0544
2000	0.0903	0.0953	0.1010
5000	0.1233	0.0958	0.2070

*Note: As we've subsampled the framerate from 120fps to 30fps, 80ms=67ms (2nd frame after our model's prediction of the 60th frame, where 0-59 frames were fed as context. This is therefore the 62nd frame) and similarly, 320ms=333ms.

C. Qualitative Visual Analysis

To provide a qualitative perspective on the model's forecasting capabilities, Fig. 2 and Fig. 3 illustrate the best-case and worst-case pose predictions generated by our autoregressive Transformer, respectively.

In the best-case scenario (Fig. 2), the predicted 3D joint locations tightly align with the ground-truth sequence. The model successfully maps the underlying kinematic dynamics from the historical context window, producing a future motion trajectory that is smooth, temporally continuous, and highly accurate even across extended horizons. This indicates the Transformer's proficiency in modeling predictable, periodic, or stable momentum-based movements.

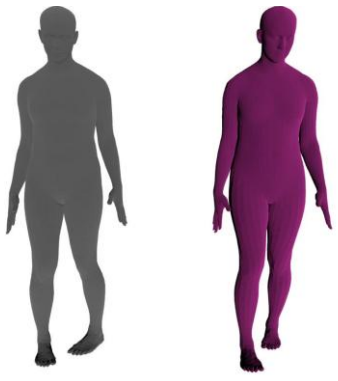


Fig. 2: Qualitative visualization of the best-case motion forecasting result. The ground-truth pose is depicted in gray, and the autoregressive Transformer prediction is shown in purple.

Fig. 3 depicts the worst-case scenario, highlighting the inherent challenges associated with long-horizon autoregressive forecasting. In this instance, the predicted trajectory visibly diverges from the ground truth. This

deviation is primarily driven by error accumulation; because the model iteratively feeds its own predicted residual velocities back into the context window, minor inaccuracies in early frames compound over longer time horizons (e.g., at 5.0 s). This phenomenon is especially pronounced when the subject performs highly stochastic, abrupt, or complex actions that fall outside a predictable distribution.

However, it is crucial to note a significant advantage of our architecture evident even in the worst-case prediction: despite the spatial deviation from the true motion path, the generated skeleton in Fig. 3 remains anatomically plausible. Because the Transformer operates strictly within the regularized VPoser latent space, it is bounded by human anatomical constraints. As a result, the model completely avoids the unnatural bone stretching, skeletal distortion, and joint intersections that frequently plague models predicting raw 3D Cartesian coordinates directly.

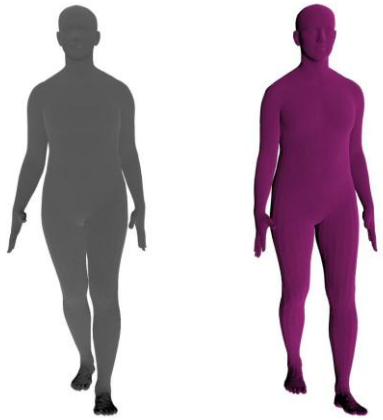


Fig. 3: Qualitative visualization of the worst-case motion forecasting result. The ground-truth pose is depicted in gray, and the autoregressive Transformer prediction is shown in purple.

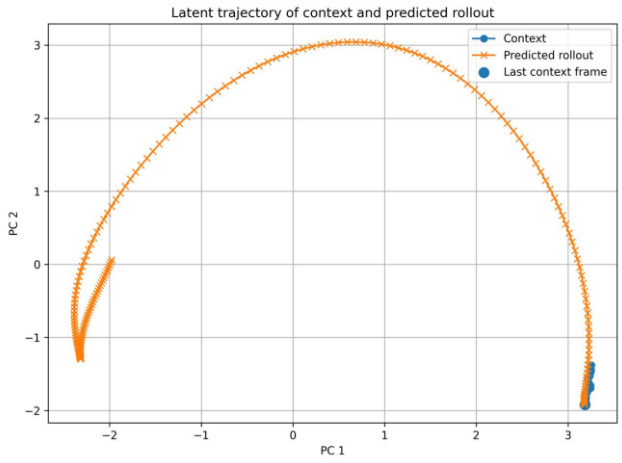


Fig. 4: PCA projection of latent trajectories showing smooth and continuous evolution during autoregressive rollout.

The PCA visualization further highlights the stability of the learned latent dynamics. Despite the accumulation of prediction error over time, the generated trajectories follow a smooth and continuous path in the reduced-dimensional space. This suggests that the model successfully captures the underlying structure of human motion, even when deviations from the ground truth occur. Such behavior reinforces the advantage of learning and forecasting within a constrained latent manifold.

IV. Limitations

Despite its strong performance, the proposed approach has several limitations. First, the autoregressive rollout mechanism leads to the accumulation of prediction errors over long horizons, as each prediction depends on previously generated outputs. This results in gradual divergence from the ground-truth motion, particularly for complex or highly stochastic actions.

Second, the model relies on a fixed latent representation obtained from a pretrained VPoser encoder. While this ensures anatomical consistency, it may limit the model's ability to capture task-specific motion nuances that are not well represented in the pretrained latent space.

Finally, the current framework does not explicitly model environmental context or human-object interactions, which are critical factors in real-world motion prediction scenarios. Addressing these limitations is essential for improving long-term forecasting accuracy and generalization.

V. Future Work

While the proposed Transformer-based approach demonstrates strong performance in short- and mid-term motion forecasting, several avenues remain for future improvement. One key limitation of the current autoregressive formulation is the accumulation of prediction error over long horizons, as observed in the qualitative and quantitative results. Future work could explore non-autoregressive or diffusion-based sequence modeling approaches to mitigate this compounding error effect.

Additionally, the current model operates on a fixed latent representation derived from a pretrained VPoser encoder. Jointly training the latent representation and the forecasting model in an end-to-end manner may improve task-specific feature learning and overall prediction accuracy. Another promising direction involves incorporating richer temporal context, such as longer history windows or hierarchical temporal modeling, to better capture complex motion patterns.

Furthermore, experimenting with different loss formulations, including varying weightings between reconstruction loss (MAE) and geometric error (MPJPE), could provide improved balance between latent accuracy and physical realism. Finally, extending the model to incorporate additional modalities, such as scene context or interaction cues, could enable more robust predictions in real-world human motion scenarios.

VI. Conclusion

In this work, we presented an autoregressive Transformer-based framework for human motion forecasting operating in a compact latent space learned via VPoser. By leveraging a structured representation of human pose, the model is able to generate smooth and anatomically plausible motion sequences while avoiding common issues such as bone stretching and geometric inconsistencies associated with direct 3D prediction.

Quantitative evaluation using MPJPE demonstrates that the proposed approach outperforms simple deterministic baselines, particularly at longer prediction horizons, indicating its ability to capture non-linear motion dynamics. Qualitative results further show that the model produces

coherent and realistic motion, even in cases where predictions deviate from the ground truth.

However, performance degrades over extended horizons due to error accumulation inherent in autoregressive rollouts. Despite this limitation, the results highlight the effectiveness of combining latent-space modeling with Transformer architectures, suggesting a promising direction for future research in long-term human motion forecasting.

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